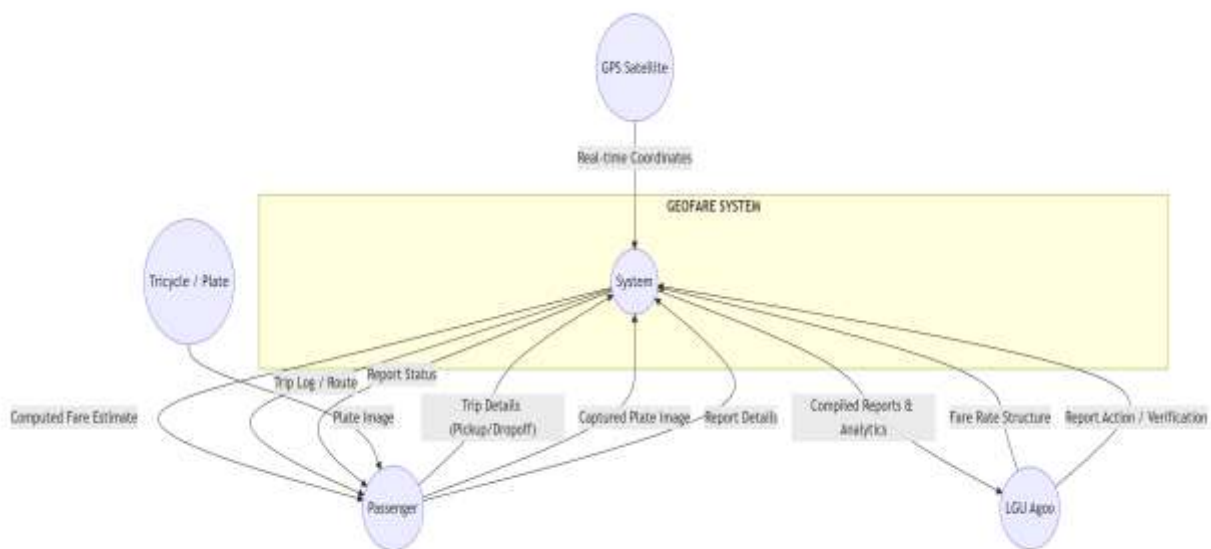


DATA FLOW DIAGRAM

GEOFARE: A Web-Based GPS Fare Computation and Route Monitoring System for Tricycle Passengers Safety in Agoo, La Union

LVLS OF DFD

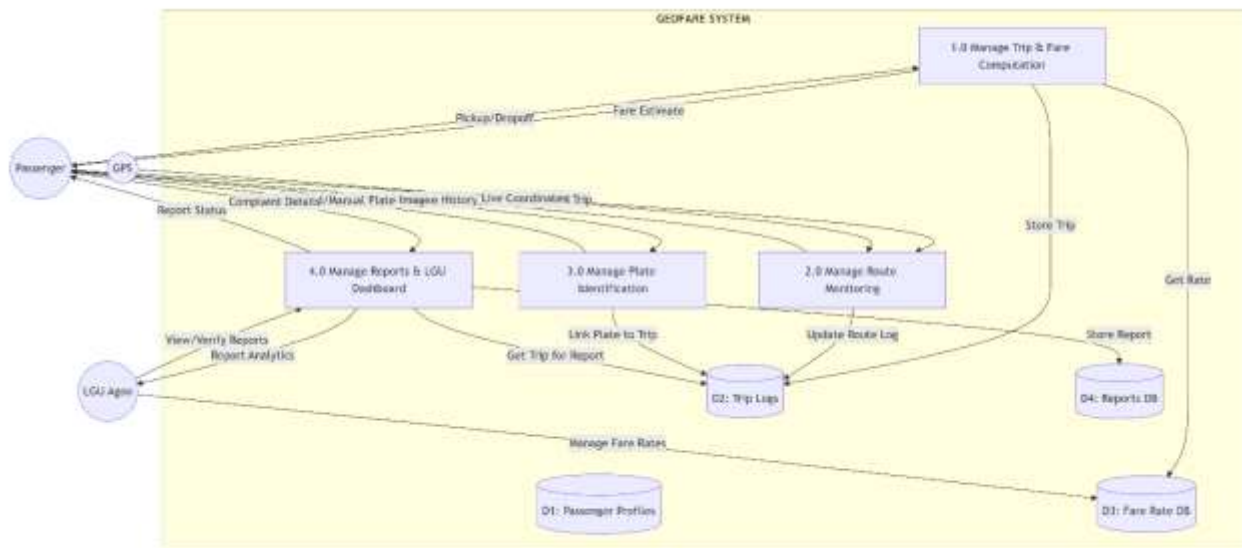
Level 0: Context Diagram



External Entities

1. **Passenger:** The primary user who requests fare estimates, logs trips, and submits reports.
2. **Tricycle (Plate Number):** The physical object/person that the passenger observes and captures an image of for identification.
3. **LGU (Agoo):** The local government unit that receives reports and manages the fare rate structure.
4. **GPS Satellite:** The external system that provides location data.

Level 1: High-Level Process



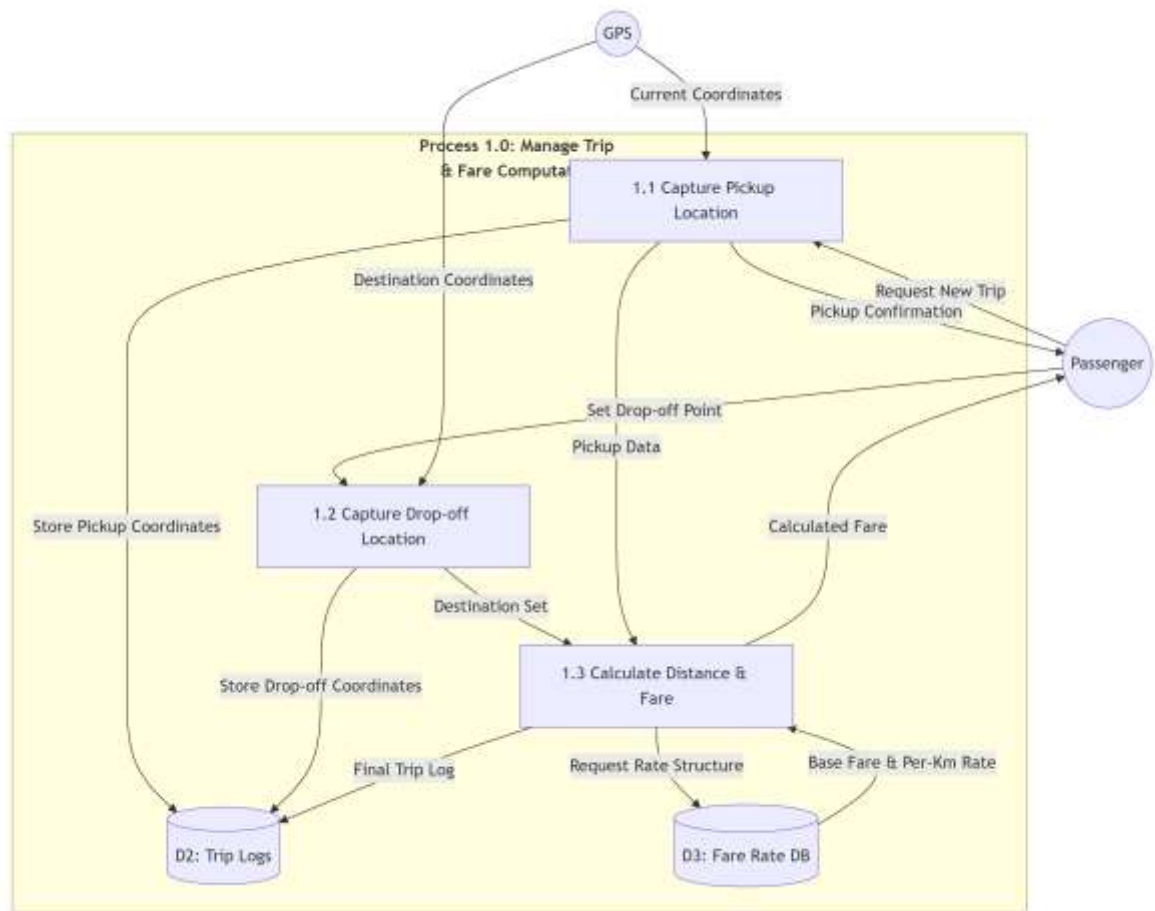
Processes:

1. **1.0 Manage Trip & Fare Computation:** Handles GPS input, distance calculation, and fare estimation.
2. **2.0 Manage Route Monitoring:** Logs the GPS path during the actual trip.
3. **3.0 Manage Plate Identification:** Handles OCR processing and manual entry for the plate number.
4. **4.0 Manage Reports & LGU Dashboard:** Handles complaint submission and the LGU's administrative interface.

Data Stores:

- **D1:** Passenger Profiles (optional, for history).
- **D2:** Trip Logs (stores all trip details, routes, and fare computations).
- **D3:** Fare Rate Database (stores the official base fare + per-km rate).
- **D4:** Reports Database (stores all submitted complaints and their status).

Level 2: Detailed Process of “1.0 Manage Trip & Fare Computation”



Sub-Processes

1. **1.1 Capture Pickup Location:** Gets the passenger's starting GPS coordinates.
2. **1.2 Capture Drop-off Location:** Gets the destination GPS coordinates.
3. **1.3 Calculate Distance & Fare:** Computes the distance and applies the official rate structure.